



# Aviatrix Cloud Firewall

SECURE EGRESS

ACE Team

# Problem Statement

## Private workloads need internet access

- SaaS integration



- Patching

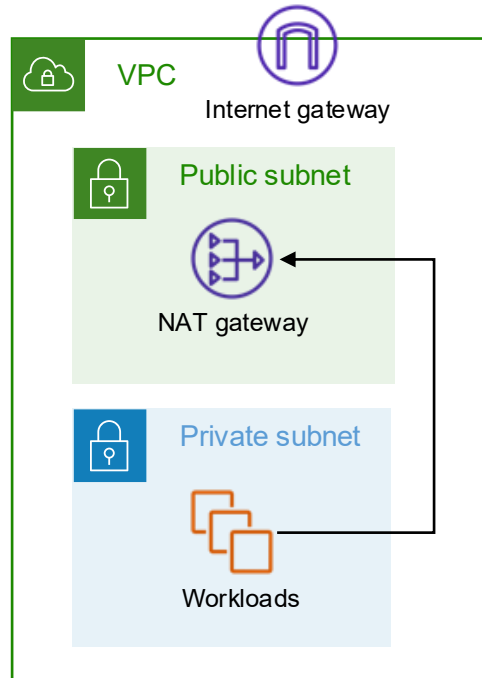


- Updates



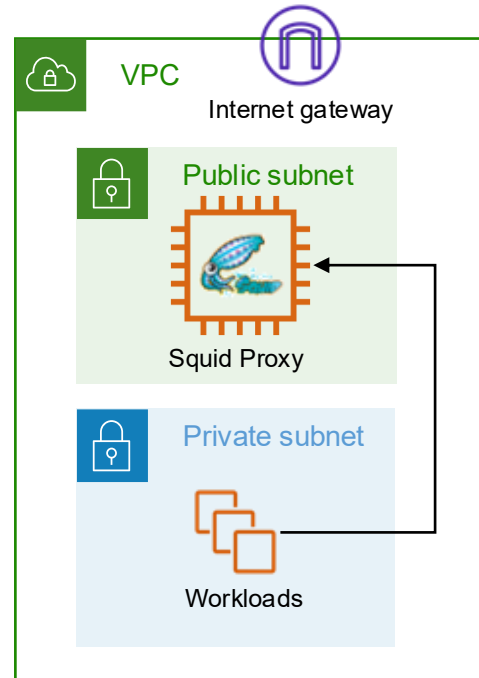
### NAT Gateway

- Layer-4 only
- NACLs management



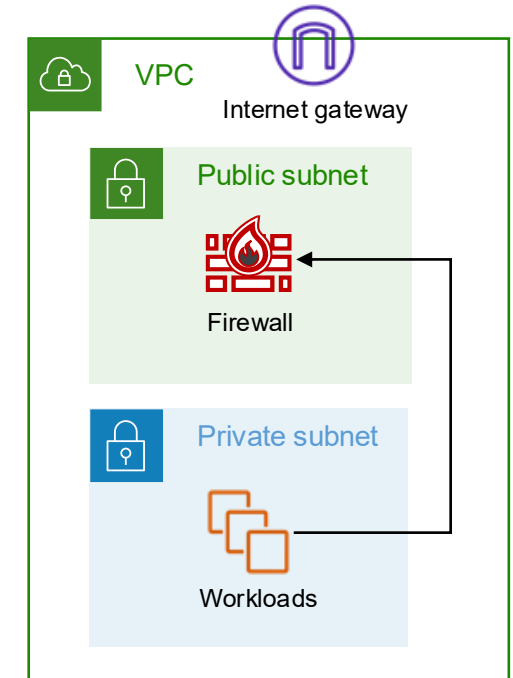
### Squid Proxy

- Hard to manage
- Scale and HA issues

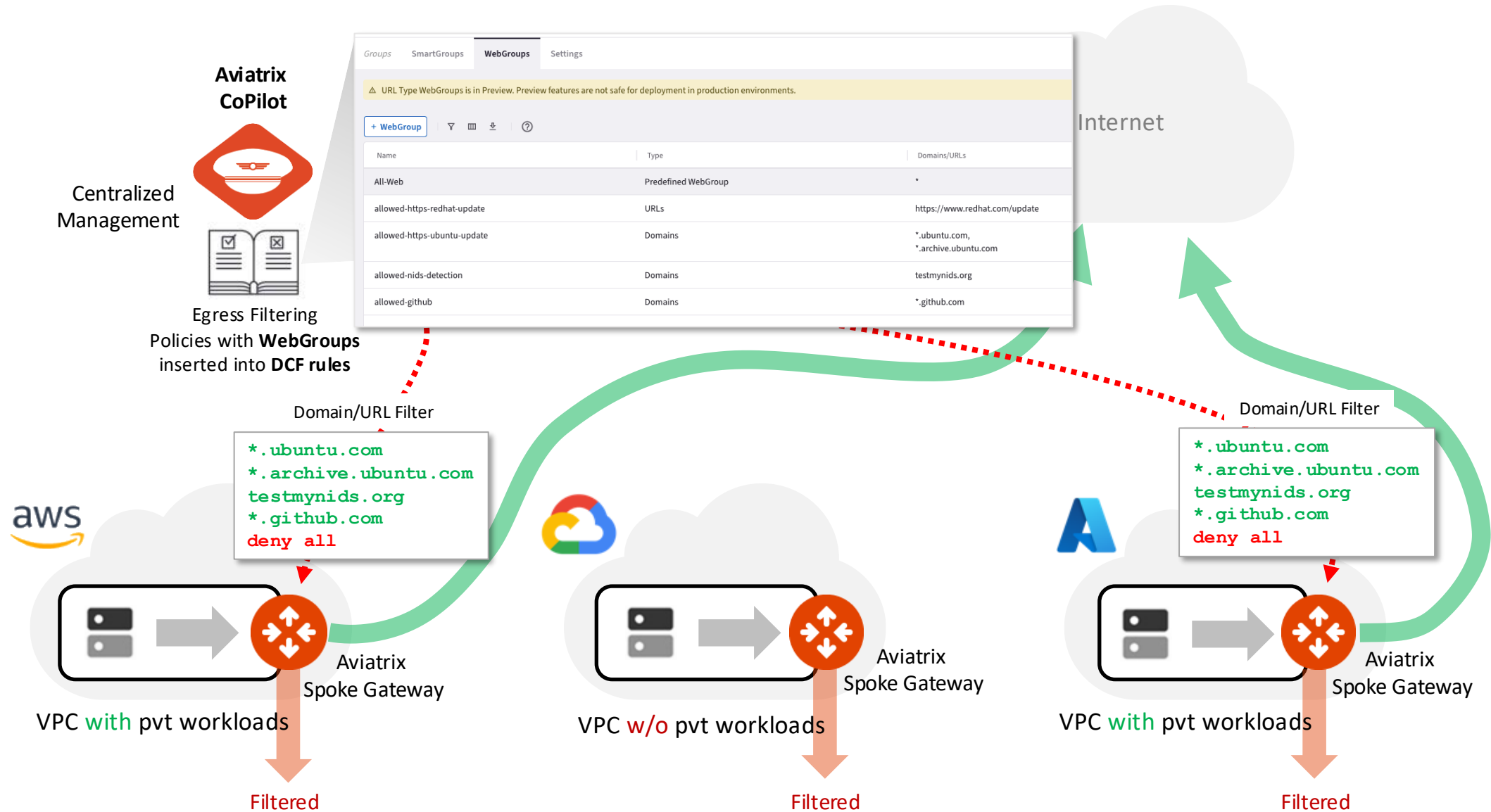


### Layer-7 Firewall

- Overkill
- Expensive



# Aviatrix Secure Egress Filtering

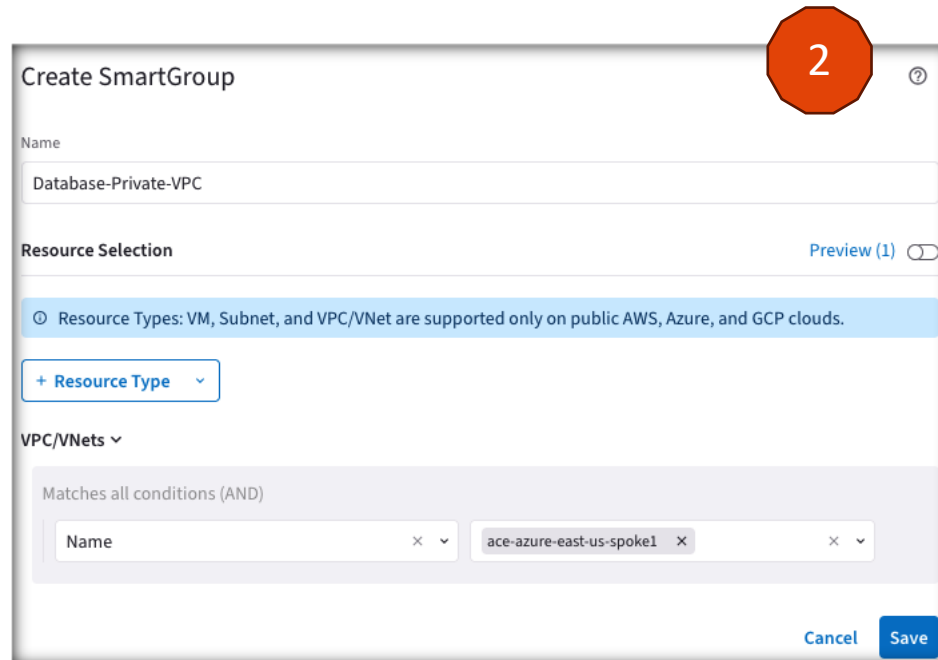
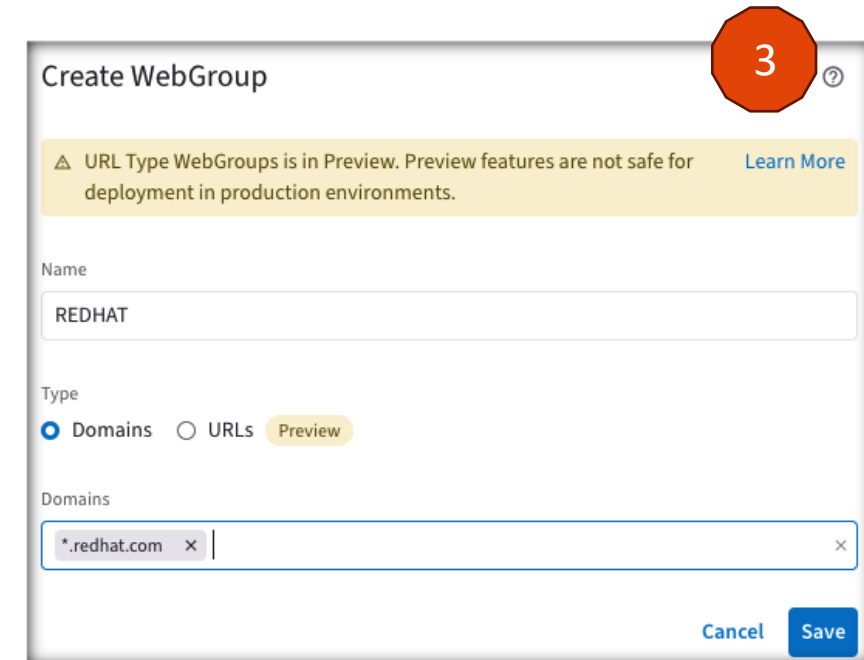




# Tools for Troubleshooting Secure Egress

# Best Practice 01 – Preparation for the DCF Policies

- Prioritize creating the fundamental elements for the Distributed Cloud Firewall policies for your egress traffic. To streamline and secure your traffic management, utilize the following grouping mechanisms:
- **SmartGroups:** Logical groupings of resources defined by specific tags or attributes (i.e. the L3 info)
- **WebGroups:** Pre-defined sets of trusted domains or URLs. (i.e. the payload/domain/url)
- **ExternalGroups:** Collections synchronized from external identity providers (e.g., GitHub or Azure Services).

2  
A form titled "Create SmartGroup". The "Name" field contains "Database-Private-VPC". The "Resource Selection" section has a toggle for "Preview (1)" and a note: "Resource Types: VM, Subnet, and VPC/VNet are supported only on public AWS, Azure, and GCP clouds." Below is a "+ Resource Type" button. The "VPC/VNets" section is expanded, showing "Matches all conditions (AND)" with a search bar containing "ace-azure-east-us-spoke1". "Cancel" and "Save" buttons are at the bottom right.3  
A form titled "Create WebGroup". A yellow warning banner states: "URL Type WebGroups is in Preview. Preview features are not safe for deployment in production environments." with a "Learn More" link. The "Name" field contains "REDHAT". The "Type" section has radio buttons for "Domains" (selected), "URLs", and a "Preview" button. The "Domains" field contains "\*.redhat.com". "Cancel" and "Save" buttons are at the bottom right.

# Best Practice 2 – Creation of the egress DCF Policy

Create Rule
1
?

**Name**

**Source Groups**

Database-Private-VPC
x
v

**Destination Groups**

Public Internet
x
v

**WebGroups**

REDHAT
x
v

**Protocol**

TCP
v

**Port**

443
x
v

Specify multiple ports (e.g. 80) and/or port ranges (e.g. 80-8080)

**Rule Behavior**

Action

Deny
v

SG Orchestration

☐ Off

TLS Decryption

☐ Off

Enforcement ☒ On
Logging ☒ On

**Rule Priority**

Place Rule

To Top
v

Cancel
Save In Drafts

- When creating dedicated Egress Policies, follow a Zero Trust approach by implementing an explicit "Deny All" rule at the bottom of your policy list. Use the following configuration steps:
- **Define Source Groups:** Populate your Source Groups with **SmartGroups** that include the specific elements—such as VOPCs and subnet workloads—that should be restricted from general internet access.
- **Apply WebGroups:** Assign the appropriate **WebGroups** to define permissible outbound traffic destinations.
- **Enforce Zero Trust:** Always conclude your policy set with an **explicit "Deny All" rule** to ensure that any traffic not explicitly permitted is automatically blocked.

Distributed Cloud Firewall
2

Ruleset: V1 Policy List Manage Rulesets

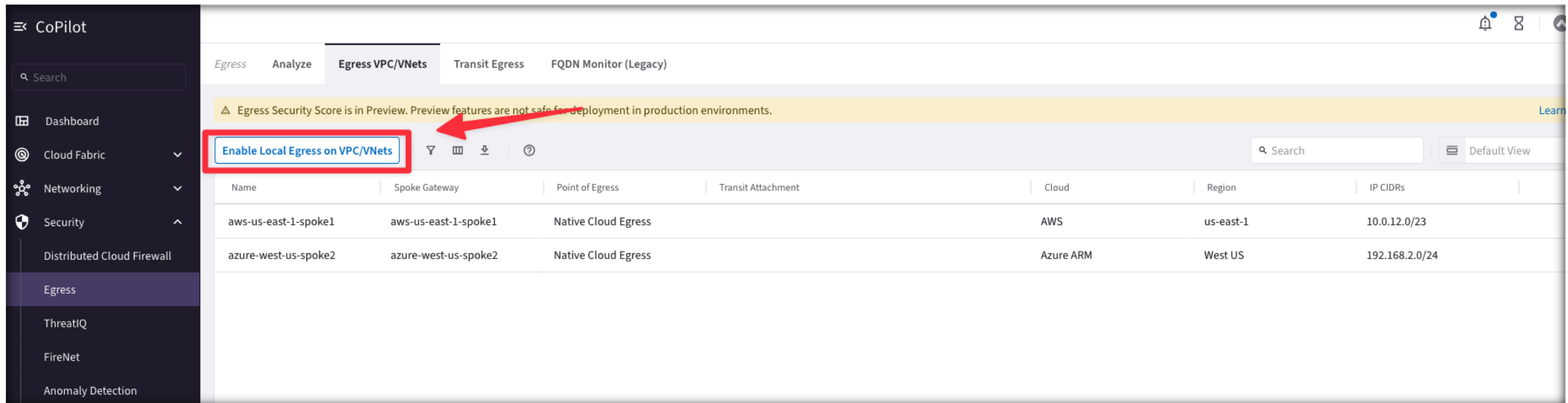
+ Rule
2 Added
Discard
Commit

| Priority | Name                  | Source               | Destination          | WebGroup | Protocol | Ports | Action | SG Orchestration |
|----------|-----------------------|----------------------|----------------------|----------|----------|-------|--------|------------------|
| 0        | Egress-for-Azure-VNet | Database-Private-VPC | Public Internet      | REDHAT   | TCP      | 443   | Deny   |                  |
| 1        | ExplicitDenyAll       | Anywhere (0.0.0.0/0) | Anywhere (0.0.0.0/0) |          | Any      |       | Deny   |                  |

Zero trust Approach = Allow List Model

# Best Practice 3 – Enable the Egress

- Enable egress on the **Spoke Gateway** only as the final step of the deployment.
- **Ensure Continuous Security:** By delaying the redirect of the default route until the policies are active, you guarantee that the Fabric is already governing all traffic generated from internal private workloads the moment they begin reaching the gateway.



The screenshot shows the Aviatrix CoPilot interface with the 'Egress VPC/VNets' tab selected. A red box highlights the 'Enable Local Egress on VPC/VNets' button. A red arrow points to this button from a yellow warning banner that reads: 'Egress Security Score is in Preview. Preview features are not safe for deployment in production environments.' Below the button is a table with the following data:

| Name                 | Spoke Gateway        | Point of Egress     | Transit Attachment | Cloud     | Region    | IP CIDRs       |
|----------------------|----------------------|---------------------|--------------------|-----------|-----------|----------------|
| aws-us-east-1-spoke1 | aws-us-east-1-spoke1 | Native Cloud Egress |                    | AWS       | us-east-1 | 10.0.12.0/23   |
| azure-west-us-spoke2 | azure-west-us-spoke2 | Native Cloud Egress |                    | Azure ARM | West US   | 192.168.2.0/24 |



Next:

## Lab 7 Aviatrix Cloud Firewall